Toric Geometry

— ***EYNTKA*** **Series** —

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Introduction

Toric geometry is the study of the algebraic varieties that a torus builds. A torus $(\mathbb{C}^*)^n$ is the simplest object in algebraic geometry that is both a variety and a group, and when it is allowed to act on a larger variety with a dense orbit, the resulting geometry is controlled completely by combinatorics: by lattices of integer points, by cones and their duals, by fans assembled from those cones, and by the lattice polytopes dual to them. The subject is, at bottom, a dictionary. On one side stand geometric objects and their properties, a variety, its smoothness or singularities, its divisors, its line bundles, the dimensions of its cohomology groups; on the other stand combinatorial objects, a cone, a fan, a polytope, a count of lattice points. The dictionary translates each side into the other, and because the combinatorial side can be drawn on paper and computed by hand, questions that are hard in general algebraic geometry become finite, visible, and often elementary here.

This is what makes toric geometry the standard laboratory of the field. Nearly every construction one meets in a first course, products and blow-ups, proper and non-proper morphisms, Weil and Cartier divisors, ample and nef cones, sheaf cohomology and Riemann-Roch, resolution of singularities, appears in the toric world in a form one can carry out completely on an example. The book follows that arc across six parts: affine toric varieties and the cones that classify the normal ones; projective toric varieties and lattice polytopes; fans and the abstract varieties glued from them, with the orbit-cone correspondence that reads a variety's torus orbits off its fan; toric divisors, the class group, and the Cox construction that presents every toric variety as a quotient; line bundles, sheaf cohomology computed by counting lattice points, and toric Riemann-Roch; and finally resolution of toric singularities and a capstone on reflexive polytopes and mirror symmetry. A recurring cast of examples, affine space, the projective spaces, $\mathbb{P}^1 \times \mathbb{P}^1$ and the Hirzebruch surfaces, weighted projective spaces, the cyclic quotient singularities, is carried through every stage, so that each new idea is met first on objects already familiar.

The book is written to serve two readers at once, and its structure reflects a deliberate choice about where they part ways. For a reader who has not yet studied schemes, the first four parts are a concrete gateway into algebraic geometry: they are developed over an algebraically closed field, by default the complex numbers, with varieties rather than schemes as the objects, and they ask for no more background than affine and projective varieties and a working command of commutative algebra. For a reader arriving from a course on schemes, the same material is a payoff, the family on which the abstract machinery becomes explicit, and the scheme-theoretic sharpenings are given where they add something, set off in boxes marked *Scheme upgrade* that a first-time reader may skip without loss. Only the fifth part, on cohomology, needs the theory of coherent sheaves, and it says so where it begins.

Prerequisites and Place in the Series

The hard prerequisites are modest. From *Classical Algebraic Geometry* [5] the reader needs affine and projective varieties over an algebraically closed field, their coordinate rings, the Zariski topology, irreducibility, morphisms, and dimension; from *Commutative Algebra* [2], finitely generated algebras and domains, prime and binomial ideals, integral closure and the characterization of normality as integral closedness, and localization. These are cited as machinery where they are used rather than re-derived. Schemes are *not* assumed for the main development. The one place the book escalates is its treatment of sheaf cohomology, which draws on coherent-sheaf cohomology from *Algebraic Geometry* [1]; that book is therefore a recommended co-requisite, read in parallel by anyone continuing to the cohomological chapters, and the optional *Scheme upgrade* boxes throughout point to it as well.

Downstream, this book is the series' toric example-bed. The Cox homogeneous-coordinate construction realizes toric varieties as the quotients that furnish the running examples of geometric invariant theory in *Moduli, Deformation Theory, and Stacks* [4]; the reflexive polytopes and Fano toric varieties of the capstone feed the mirror-symmetry and Hodge-number computations of *Hodge Theory* [3]; and the toric intersection theory sketched at the end connects to the intersection-theoretic thread of the algebraic-geometry volumes. A reader may enter here directly after classical algebraic geometry and commutative algebra, well before schemes, and return to the later chapters once the cohomological machinery is in hand.

Part I

Affine Toric Varieties and the Cone Dictionary

The subject begins locally. Before a global toric variety can be assembled, one needs the affine pieces from which it is glued, and this Part builds them and the combinatorial data that classifies them. The organizing discovery is that a single affine toric variety can be described in four interchangeable languages: as a variety carrying a dense torus whose action extends, as the closure of the image of a monomial map, as the zero locus of a binomial (toric) ideal, and as the spectrum of a semigroup algebra. That these four descriptions coincide is the first theorem of the theory, and it is what makes toric geometry computable: every geometric question can be pushed into whichever language answers it most cleanly.

The dictionary then acquires its geometric half. Among all affine toric varieties the *normal* ones are singled out by a clean combinatorial object, a strongly convex rational polyhedral cone, and the correspondence between such cones and normal affine toric varieties, together with the reading of smoothness, dimension, and singularity type directly off the cone, is the payoff of the Part. Everything here is developed over an algebraically closed field, by default the complex numbers, so that the torus $(\mathbb{C}^*)^n$ carries its familiar analytic topology; the reader who has met affine varieties and their coordinate rings has all that is required.

1

Tori, Characters, and Lattices

Every toric variety is built from a single torus, and the torus is built from a lattice. This chapter fixes that ground floor. A torus $(\mathbb{C}^*)^n$ looks at first like an analytic object, homotopy-equivalent to the compact torus $(S^1)^n$, but its algebraic life is governed entirely by two lattices of integer data: the characters χ^m, which are the group homomorphisms $T \to \mathbb{C}^*$, and the one-parameter subgroups λ^u, which are the group homomorphisms $\mathbb{C}^* \to T$. Each collection turns out to be a free abelian group of rank n, and the two are dual.

The payoff is a dictionary. A torus determines, and is determined by, its lattice of characters together with the dual lattice of one-parameter subgroups; a morphism of tori is nothing more than a $\mathbb{Z}$-linear map of lattices. This equivalence between tori and lattices is what makes the entire subject combinatorial: from this chapter onward, geometric constructions on tori and the varieties built from them will be read off integer-linear algebra. We begin with the torus itself, then extract its two lattices, and close by pairing them.

1.1 The Algebraic Torus

Every object in this book is assembled from one primitive piece: a copy of the multiplicative group $\mathbb{C}^*$, taken to some power and glued to its neighbours by monomial data. The whole subject is the study of how far one can push this single building block, so it pays to understand it thoroughly before anything is built on top of it. The algebraic torus is the geometric form of a lattice, and the dictionary between the two runs through the entire theory. This first section isolates the torus as an object in its own right: what it is, when two of them are "the same," and which maps between them the theory uses.

The name *torus* is borrowed from topology, and the borrowing is worth unpacking at the outset because it explains a choice of vocabulary that would otherwise look arbitrary. A single copy of $\mathbb{C}^*$ deformation-retracts onto the unit circle S^1, so a product of n copies retracts onto the n-fold product $(S^1)^n$, which is the ordinary topological n-torus. The algebraic object $(\mathbb{C}^*)^n$ is thus a complex-geometric thickening of the compact torus one meets in elementary topology, and much of the intuition (a torus is a product of circles, it has a natural group structure, its "directions" are indexed by integers) carries over verbatim. That intuition is a reliable guide, and it will be made precise in example 1.1.5.

Fix an algebraically closed field k; throughout the book $k = \mathbb{C}$ unless stated otherwise, and the reader may safely read every "k" below as "$\mathbb{C}$." The starting point is the multiplicative group of the field, which is the smallest nontrivial torus and the atom from which all the rest are built.

Definition 1.1.1: The Multiplicative Group $\mathbb{G}_m$

The *multiplicative group* is the affine variety

$$\mathbb{G}_m := \mathbb{C}^* = \mathbb{C} \setminus \{0\}$$

equipped with multiplication $(s,t) \mapsto st$ as its group law and 1 as its identity. As a variety it is the affine open subset of $\mathbb{A}^1$ where the coordinate is invertible, and as a ring of functions it is the Laurent polynomial ring $\mathbb{C}[x, x^{-1}] = \mathbb{C}[x^{\pm 1}]$.

Two facts about $\mathbb{G}_m$ deserve to be stated plainly, because both are used constantly and both are places where the geometry and the algebra meet. First, $\mathbb{G}_m$ is an affine variety: the locus $\{(x,y) \in \mathbb{A}^2 \middle| xy = 1\}$ is Zariski closed in $\mathbb{A}^2$, and projection to the x-coordinate identifies it with $\mathbb{C}^*$, so $\mathbb{C}^*$ inherits the structure of an affine variety with coordinate ring $\mathbb{C}[x,y]/(xy-1) \cong \mathbb{C}[x^{\pm 1}]$. The passage from the open subset $\mathbb{C} \setminus \{0\}$ of $\mathbb{A}^1$ to a closed subvariety of $\mathbb{A}^2$ is the standard device for exhibiting a principal open set as an affine variety, and it is developed in full in [5]. Second, the group law st and the inversion $t \mapsto t^{-1}$ are morphisms of varieties, since in the coordinates (x,y) with $xy = 1$ multiplication is polynomial and inversion is the coordinate swap $(x,y) \mapsto (y,x)$. A variety whose group operations are morphisms is what one means by an *algebraic group*, and $\mathbb{G}_m$ is the prototype.

The torus in n variables is the n-fold product of this atom, with the group law applied slot by slot. Making the product literal, rather than an abstract categorical product, is what lets one compute with tori by hand.

Definition 1.1.2: The Standard Torus

The *standard n-dimensional torus* is the product variety

$$(\mathbb{C}^*)^n = \underbrace{\mathbb{C}^* \times \cdots \times \mathbb{C}^*}_{n}$$

with *component-wise multiplication*

$$(s_1, \dots, s_n) \cdot (t_1, \dots, t_n) = (s_1 t_1, \dots, s_n t_n)$$

and identity $(1, \dots, 1)$. Its coordinate ring is the Laurent polynomial ring in n variables,

$$\mathbb{C}[(\mathbb{C}^*)^n] = \mathbb{C}[x_1^{\pm 1}, \dots, x_n^{\pm 1}]$$

so that $(\mathbb{C}^*)^n = \operatorname{mSpec} \mathbb{C}[x_1^{\pm 1}, \dots, x_n^{\pm 1}]$.

The identification of the coordinate ring is the hinge of the whole subject and should be dwelt on. The ring $\mathbb{C}[x_1^{\pm 1}, \dots, x_n^{\pm 1}]$ is the localization of the polynomial ring $\mathbb{C}[x_1, \dots, x_n]$ at the product $x_1 \cdots x_n$; its elements are $\mathbb{C}$-linear combinations of *Laurent monomials* $x_1^{a_1} \cdots x_n^{a_n}$ with the exponent vector $(a_1, \dots, a_n)$ ranging over all of $\mathbb{Z}^n$, not just $\mathbb{N}^n$. Localization at a single element is a standard construction of commutative algebra, and the resulting ring is a finitely generated $\mathbb{C}$-algebra and an integral domain; both facts are recorded in [2]. Its maximal spectrum mSpec , the set of closed points of the associated affine variety, is exactly $(\mathbb{C}^*)^n$: a Laurent monomial is invertible precisely off the coordinate hyperplanes, so imposing invertibility of each coordinate deletes those hyperplanes from $\mathbb{A}^n$ and leaves $(\mathbb{C}^*)^n$. The lattice $\mathbb{Z}^n$ of exponent vectors will reappear in the next section as the *character lattice*, and the fact that monomials are indexed by a lattice while the variety is a product of $\mathbb{C}^*$'s is the first line of the combinatorics-geometry dictionary.

An abstract torus is any variety that looks like a standard torus, where "looks like" must respect *both* structures the standard torus carries: it is a variety and it is a group, and an isomorphism of tori is required to preserve both at once. Insisting on both is not pedantry. A bare isomorphism of varieties could scramble the group law, and a bare isomorphism of abstract groups need not be algebraic at all, so only the combined notion captures the object the theory needs.

Definition 1.1.3: Algebraic Torus

An *algebraic torus* (or simply a *torus*) is an affine variety T over $\mathbb{C}$ together with a group structure on its set of points, such that there exists an isomorphism

$$\varphi \colon T \xrightarrow{\sim} (\mathbb{C}^*)^n$$

that is *simultaneously an isomorphism of varieties and an isomorphism of groups*, where $(\mathbb{C}^*)^n$ carries component-wise multiplication (definition 1.1.2). The integer n is the *dimension* of T, written $\dim T = n$.

The dimension is well defined: an isomorphism of varieties preserves the dimension of the underlying variety, and $\dim(\mathbb{C}^*)^n = n$ because $(\mathbb{C}^*)^n$ is a dense open subset of $\mathbb{A}^n$, which has dimension n; the invariance of dimension under isomorphism of varieties is proved in [5]. So two tori of different dimensions can never be isomorphic, and n is an invariant of T. The requirement that φ be an isomorphism of groups as well as of varieties means that T is automatically an algebraic group: transporting the morphisms μ (multiplication) and ι (inversion) of $(\mathbb{C}^*)^n$ along φ makes the group operations on T into morphisms of varieties, since φ and φ^{-1} are morphisms. A torus is therefore a commutative algebraic group, and the qualifier "algebraic" distinguishes it from the compact real torus $(S^1)^n$, which is a Lie group but not an algebraic variety over $\mathbb{C}$.

One subtlety is worth flagging before any examples. The definition asserts only that *some* isomorphism φ to a standard torus exists; it does not hand one over, and different choices of φ differ by an automorphism of $(\mathbb{C}^*)^n$. Automorphisms of $(\mathbb{C}^*)^n$ correspond to invertible integer matrices $\mathrm{GL}_n(\mathbb{Z})$ acting on exponent vectors, a fact proved in section 1.3, so an abstract torus carries no preferred set of coordinates. This coordinate-freeness is exactly what makes the lattice viewpoint of the coming sections the right one: the intrinsic data of a torus is not a choice of coordinates but the lattice of characters, on which $\mathrm{GL}_n(\mathbb{Z})$ acts as change of basis.

Having said which objects are tori, it remains to say which maps between them are allowed. As with the objects, the correct notion of morphism respects both structures at once: a map of tori is required to be a morphism of varieties and a homomorphism of groups simultaneously. These are the arrows in the category of tori, and the entire lattice dictionary of section 1.3 is a statement about them.

Definition 1.1.4: Toric Morphism

Let T_1 and T_2 be algebraic tori. A *toric morphism* (or *morphism of tori*) is a map $\varphi\colon T_1 \to T_2$ that is *simultaneously*:

1. a morphism of varieties, and
2. a homomorphism of the underlying groups, that is, $\varphi(s \cdot t) = \varphi(s) \cdot \varphi(t)$ for all $s, t \in T_1$.

A toric morphism that is also an isomorphism of varieties with a toric inverse is an *isomorphism of tori*.

The two conditions pull in different directions, and combining them is what makes toric morphisms rigid enough to be classified. A morphism of varieties $\varphi\colon (\mathbb{C}^*)^m \to (\mathbb{C}^*)^n$ is given in coordinates by n invertible regular functions on $(\mathbb{C}^*)^m$, that is, by n Laurent monomials up to nonzero scalars. The homomorphism condition forces each of those scalars to be 1 and each function to be a monomial (a Laurent monomial evaluated at the identity must give the identity 1, which kills the scalar). A toric morphism is therefore given by a tuple of monomials with integer exponents, and the exponent data is precisely an integer matrix. Chasing this observation to its conclusion is the business of section 1.2 and section 1.3, where it becomes the statement that toric morphisms $T_1 \to T_2$ are the same thing as $\mathbb{Z}$-linear maps between the character lattices; for now the point is only that requiring both structures at once is what makes the eventual classification possible.

A concrete case makes the rigidity visible. The squaring map $\varphi\colon \mathbb{C}^* \to \mathbb{C}^*$, $t \mapsto t^2$, is a toric morphism: it is a morphism of varieties (given by the monomial x^2) and a group homomorphism. Its exponent is the integer 2, and one can already see the lattice map $\mathbb{Z} \to \mathbb{Z}$, multiplication by 2, lurking underneath. By contrast the affine map $t \mapsto t+1$ is a morphism of varieties $\mathbb{C}^* \to \mathbb{C}$ but sends 1 to 2, not to the identity, so it is not a homomorphism and hence not a toric morphism; and complex conjugation $t \mapsto \bar{t}$ is a group automorphism of $\mathbb{C}^*$ that is not a morphism of varieties. Neither map meets the requirement that both structures be respected, which is exactly the point of imposing both.

The definitions are now in place, and it remains to see them at work on the objects that will recur throughout the book. The following example collects the standard tori, exhibits a torus sitting inside a larger one, and makes the analytic picture $(S^1)^n$ precise.

Example 1.1.5: The Standard Tori and a Subtorus

1. *The one-dimensional torus* $\mathbb{G}_m = \mathbb{C}^*$. Here $n = 1$. The variety is $\mathbb{C} \setminus \{0\}$, the group law is ordinary multiplication of nonzero complex numbers, and the coordinate ring is $\mathbb{C}[x^{\pm 1}]$. This is the atom: $\dim \mathbb{G}_m = 1$, and every torus is built from copies of it.

2. *The standard torus* $(\mathbb{C}^*)^n$. By definition 1.1.2 this is a torus with the identity isomorphism $\varphi = \mathrm{id}$; its dimension is n and its coordinate ring is $\mathbb{C}[x_1^{\pm 1}, \ldots, x_n^{\pm 1}]$. Concretely, for $n = 2$ the torus $(\mathbb{C}^*)^2$ is the plane $\mathbb{A}^2$ with the two coordinate axes $\{x_1 = 0\}$ and $\{x_2 = 0\}$ removed, and multiplication is $(s_1, s_2)(t_1, t_2) = (s_1t_1, s_2t_2)$.

3. *A subtorus of* $(\mathbb{C}^*)^2$. Consider the diagonal image of the map
$$\lambda\colon \mathbb{C}^* \longrightarrow (\mathbb{C}^*)^2, \qquad t \longmapsto (t, t)$$
This λ is a toric morphism: it is given by the monomials (x, x), so it is a morphism of varieties, and $\lambda(st) = (st, st) = (s, s)(t, t) = \lambda(s)\lambda(t)$, so it is a group homomorphism. Its image is $\Delta = \{(t, t) \big| t \in \mathbb{C}^*\}$, the diagonal, which is a Zariski-closed subgroup of $(\mathbb{C}^*)^2$ (it is cut out by the equation $x_1 = x_2$) and is isomorphic to $\mathbb{C}^*$ as a torus via λ. Thus Δ is a one-dimensional *subtorus* of the two-dimensional torus $(\mathbb{C}^*)^2$: a Zariski-closed subgroup that is itself a torus. More generally, for any integer a the map $t \mapsto (t, t^a)$ has image a one-dimensional subtorus, and the exponent a is the seed of the lattice map $\mathbb{Z} \to \mathbb{Z}^2$, $1 \mapsto (1, a)$, that classifies it.

4. *The analytic picture* $(S^1)^n$. Writing each coordinate in polar form $t_j = r_j e^{i\theta_j}$ with $r_j > 0$, the map $(r_1 e^{i\theta_1}, \ldots, r_n e^{i\theta_n}) \mapsto (e^{i\theta_1}, \ldots, e^{i\theta_n})$ deformation-retracts $(\mathbb{C}^*)^n$ onto the compact real torus
$$(S^1)^n = \{(u_1, \ldots, u_n) \in \mathbb{C}^n \big| |u_j| = 1 \text{ for all } j\}$$
by shrinking each radius r_j to 1 along the path r_j^{1-s}, $s \in [0, 1]$. The compact torus $(S^1)^n$ is a subgroup of $(\mathbb{C}^*)^n$ under component-wise multiplication, and the retraction is a homotopy equivalence. This is the origin of the word "torus": the underlying homotopy type of $(\mathbb{C}^*)^n$ is the ordinary n-torus $(S^1)^n$, the n-fold product of circles. For $n = 1$ the picture is $\mathbb{C}^*$ retracting onto the unit circle S^1, and for $n = 2$ onto the familiar doughnut surface $S^1 \times S^1$.

Two threads from this example run through the rest of the book. The subtorus in part (3) was pinned down by an integer exponent, and the analytic torus in part (4) records the same integers as winding numbers of loops; both are avatars of the lattice $\mathbb{Z}^n$ that indexes Laurent monomials. Everything that follows makes this correspondence exact, turning the informal observation "a torus is controlled by a lattice of integers" into the equivalence of categories of section 1.3. Before that can happen, the integers must be named and organized, which is the task of the character and cocharacter lattices in section 1.2.

Exercise 1.1.1

1. Verify directly that $\mathbb{C}^*$ is an affine variety by showing that the map $\mathbb{C}^* \to V(xy - 1) \subseteq \mathbb{A}^2$, $t \mapsto (t, t^{-1})$, is an isomorphism of varieties onto the closed subvariety $V(xy - 1) = \{(x, y) \big| xy = 1\}$.

Identify the coordinate ring $\mathbb{C}[x,y]/(xy-1)$ with $\mathbb{C}[x^{\pm 1}]$ explicitly, and describe the inverse morphism.

2. Show that the inversion map $\iota\colon (\mathbb{C}^*)^n \to (\mathbb{C}^*)^n$, $(t_1,\dots,t_n) \mapsto (t_1^{-1},\dots,t_n^{-1})$, is a toric morphism. Is it an isomorphism of tori? Which $\mathbb{Z}$-linear map $\mathbb{Z}^n \to \mathbb{Z}^n$ do you expect it to correspond to on exponent vectors?

3. Let $\varphi\colon (\mathbb{C}^*)^2 \to (\mathbb{C}^*)^2$ be the map $\varphi(t_1,t_2) = (t_1^2 t_2,\ t_1 t_2^3)$. Show that φ is a toric morphism and write down the 2×2 integer matrix that records its monomial exponents. Is φ an isomorphism of tori? (Hint: consider the determinant of that matrix.)

4. Show that the map $\psi\colon \mathbb{C}^* \to (\mathbb{C}^*)^2$, $t \mapsto (t,1)$, is a toric morphism whose image is a subtorus isomorphic to $\mathbb{C}^*$, and that the map $t \mapsto (t,2)$ is *not* a toric morphism. Explain in one sentence why the second map fails, referring to the group law.

5. Show that the compact torus $(S^1)^n$ is a subgroup of $(\mathbb{C}^*)^n$ under component-wise multiplication but is *not* a subvariety, that is, not the zero set of any collection of Laurent polynomials in $\mathbb{C}[x_1^{\pm 1},\dots,x_n^{\pm 1}]$. (Hint: a nonzero Laurent polynomial vanishing on the infinite set $S^1 \subseteq \mathbb{C}^*$ would have infinitely many roots.)

6. Prove that every toric morphism $\varphi\colon \mathbb{C}^* \to \mathbb{C}^*$ has the form $\varphi(t) = t^a$ for a unique integer $a \in \mathbb{Z}$. (Hint: a morphism of varieties $\mathbb{C}^* \to \mathbb{C}^*$ is an invertible regular function on $\mathbb{C}^*$, hence $c\,t^a$ for some $c \in \mathbb{C}^*$ and $a \in \mathbb{Z}$; then impose the homomorphism condition to pin down c.)

1.2 Characters and One-Parameter Subgroups

A torus T is an abstract algebraic group, and abstract groups reveal themselves through their homomorphisms. For a torus the most informative homomorphisms are those into and out of the simplest torus of all, the one-dimensional torus $\mathbb{G}_m = \mathbb{C}^*$ of definition 1.1.3. Mapping T *into* $\mathbb{G}_m$ records the ways T can act by scaling a single coordinate; mapping $\mathbb{G}_m$ *into* T records the one-parameter families of group elements that sweep through T. These two collections, the characters and the one-parameter subgroups, are the twin lattices on which the entire combinatorics of toric geometry is built. This section defines them, and then computes them completely for the standard torus $(\mathbb{C}^*)^n$: both turn out to be free abelian groups of rank n, and every character is a Laurent monomial. That computation is the first entry in the dictionary between the geometry of tori and the combinatorics of lattices.

The word "lattice" is used here in its algebraic sense: a finitely generated free abelian group, a copy of $\mathbb{Z}^n$. The reason a torus produces a lattice rather than just an abelian group is exactly the rigidity of the objects $\mathrm{Hom}(T,\mathbb{G}_m)$ and $\mathrm{Hom}(\mathbb{G}_m,T)$, which the main proposition of this section makes precise.

Definition 1.2.1: Character and the Character Lattice

Let T be an algebraic torus. A *character* of T is a toric morphism $\chi\colon T \to \mathbb{G}_m$ (definition 1.1.4), that is, a map that is simultaneously a morphism of varieties and a homomorphism of groups. The set of all characters of T, made into a group under pointwise multiplication

$$(\chi \cdot \chi')(t) = \chi(t)\,\chi'(t)$$

is called the *character lattice* of T and is written

$$M = M(T) = \operatorname{Hom}(T, \mathbb{G}_m)$$

the Hom being taken in the category of tori with toric morphisms. Its elements are indexed by an abstract lattice: a character corresponding to a lattice element $m \in M$ is denoted χ^m, so that the group law reads $\chi^m \cdot \chi^{m'} = \chi^{m+m'}$ and χ^0 is the constant character 1.

The notation χ^m is chosen so that the exponent lives additively in M while the character multiplies. Writing the group law of characters as addition of exponents is the same bookkeeping one uses for monomials: $x^a \cdot x^b = x^{a+b}$. Proposition 1.2.3 will show that for $T = (\mathbb{C}^*)^n$ the characters are *literally* the Laurent monomials, with m the exponent vector. The map $m \mapsto \chi^m$ is a group isomorphism from the additive group M to the multiplicative group of characters; the two are identified throughout, and it is harmless to speak of m itself as a character.

Dual to mapping T into $\mathbb{G}_m$ is mapping $\mathbb{G}_m$ into T. Where a character extracts a coordinate on which T scales, a homomorphism $\mathbb{G}_m \to T$ traces out a one-parameter family $t \mapsto \lambda(t)$ of elements of T, a curve through the identity that respects the group law. These are the infinitesimal directions in which one can move inside the torus.

Definition 1.2.2: One-Parameter Subgroup and the Cocharacter Lattice

Let T be an algebraic torus. A *one-parameter subgroup* (or *cocharacter*) of T is a toric morphism $\lambda\colon \mathbb{G}_m \to T$, again a map that is at once a morphism of varieties and a group homomorphism. The set of all one-parameter subgroups, made into a group under pointwise multiplication in the target,

$$(\lambda \cdot \lambda')(t) = \lambda(t)\,\lambda'(t)$$

is called the *cocharacter lattice* (or lattice of one-parameter subgroups) of T and is written

$$N = N(T) = \operatorname{Hom}(\mathbb{G}_m, T)$$

Its elements are indexed by a lattice dual to M: the one-parameter subgroup corresponding to $u \in N$ is denoted λ^u, with $\lambda^u \cdot \lambda^{u'} = \lambda^{u+u'}$ and λ^0 the constant map at the identity of T.

The two definitions are deliberately parallel, and the symmetry is the point: characters map out of T, cocharacters map into T, and the whole theory turns on the interplay between the two. The pointwise product in definition 1.2.2 makes sense only because the target T is abelian, so a product of two group homomorphisms into T is again a homomorphism; this is exactly where the commutativity of a torus is used. The letters are traditional: M (for characters) and N (for cocharacters) are the standard notation throughout toric geometry, and the pairing that will link them in section 1.3 is what justifies calling N the *dual* lattice.

Both M and N are defined for any torus T without reference to coordinates. To see that they deserve the name "lattice" one must compute them, and the natural place to compute is the standard torus $(\mathbb{C}^*)^n$, where coordinates are available. The following proposition is the foundational calculation of the chapter: it identifies every character and every one-parameter subgroup of $(\mathbb{C}^*)^n$ explicitly, and thereby shows M and N are free abelian of rank n.

Proposition 1.2.3: Characters and Cocharacters of $(\mathbb{C}^*)^n$

Let $T = (\mathbb{C}^*)^n$ with coordinates $t = (t_1, \dots, t_n)$.

1. Every character of T is a Laurent monomial: for each $m = (m_1, \dots, m_n) \in \mathbb{Z}^n$ the map

$$\chi^m : (\mathbb{C}^*)^n \to \mathbb{C}^*, \qquad \chi^m(t) = t_1^{m_1} t_2^{m_2} \cdots t_n^{m_n}$$

   is a character, and every character of T arises this way from a unique $m \in \mathbb{Z}^n$. The assignment $m \mapsto \chi^m$ is a group isomorphism $\mathbb{Z}^n \xrightarrow{\sim} M$, so $M \cong \mathbb{Z}^n$.

2. Every one-parameter subgroup of T is a monomial curve: for each $u = (u_1, \dots, u_n) \in \mathbb{Z}^n$ the map

$$\lambda^u : \mathbb{C}^* \to (\mathbb{C}^*)^n, \qquad \lambda^u(t) = (t^{u_1}, t^{u_2}, \dots, t^{u_n})$$

   is a one-parameter subgroup, and every one-parameter subgroup of T arises this way from a unique $u \in \mathbb{Z}^n$. The assignment $u \mapsto \lambda^u$ is a group isomorphism $\mathbb{Z}^n \xrightarrow{\sim} N$, so $N \cong \mathbb{Z}^n$.

In particular M and N are free abelian groups of rank $n = \dim T$.

***Proof*:**
The two parts are proved separately; each has the same three moves, produce the claimed maps, check they are toric morphisms, and show there are no others.

1. *Step 1: The Laurent monomials are characters.* Fix $m = (m_1, \dots, m_n) \in \mathbb{Z}^n$ and set $\chi^m(t) = t_1^{m_1} \cdots t_n^{m_n}$. Because each $t_i \in \mathbb{C}^*$ is invertible, negative exponents are allowed and $\chi^m(t) \in \mathbb{C}^*$, so χ^m maps $(\mathbb{C}^*)^n$ to $\mathbb{C}^*$. It is a morphism of varieties: its single coordinate function is the Laurent monomial $t_1^{m_1} \cdots t_n^{m_n}$, an element of the coordinate ring $\mathbb{C}[t_1^{\pm 1}, \dots, t_n^{\pm 1}]$ of $(\mathbb{C}^*)^n$, and its inverse $t_1^{-m_1} \cdots t_n^{-m_n}$ is again regular, so the map lands in $\mathbb{C}^*$ regularly. It is a homomorphism because exponents add:

$$\chi^m(st) = \prod_{i=1}^{n} (s_i t_i)^{m_i} = \Big(\prod_{i=1}^{n} s_i^{m_i}\Big)\Big(\prod_{i=1}^{n} t_i^{m_i}\Big) = \chi^m(s)\,\chi^m(t)$$

   where $st = (s_1 t_1, \dots, s_n t_n)$ is the component-wise product. Thus χ^m is a toric morphism, that is, a character.

   Step 2: Distinct exponent vectors give distinct characters. If $\chi^m = \chi^{m'}$ as functions, then evaluating at the standard one-parameter families isolates each exponent. For the j-th standard coordinate direction, take the point $t = (1, \dots, 1, s, 1, \dots, 1)$ with s in the j-th slot. Then $\chi^m(t) = s^{m_j}$ and $\chi^{m'}(t) = s^{m'_j}$, so $s^{m_j} = s^{m'_j}$ for all $s \in \mathbb{C}^*$. A nonzero rational function of s that is identically 1 forces $m_j = m'_j$, since $s^{m_j - m'_j} = 1$ for all $s \in \mathbb{C}^*$ has only the solution $m_j - m'_j = 0$ (a nonconstant Laurent monomial is nonconstant). As j

was arbitrary, $m = m'$. Hence $m \mapsto \chi^m$ is injective, and it is a homomorphism because $\chi^m \chi^{m'} = \chi^{m+m'}$ by the exponent-addition computation of Step 1.

Step 3: Every character is a Laurent monomial. Let $\chi\colon (\mathbb{C}^*)^n \to \mathbb{C}^*$ be any character. As a morphism of the affine variety $(\mathbb{C}^*)^n = \operatorname{mSpec} \mathbb{C}[t_1^{\pm 1}, \dots, t_n^{\pm 1}]$ to the variety $\mathbb{C}^* = \operatorname{mSpec} \mathbb{C}[u^{\pm 1}]$, the map χ corresponds to a $\mathbb{C}$-algebra homomorphism

$$\chi^\sharp\colon \mathbb{C}[u^{\pm 1}] \longrightarrow \mathbb{C}[t_1^{\pm 1}, \dots, t_n^{\pm 1}]$$

determined by the image $f := \chi^\sharp(u)$, which must be a *unit* of the Laurent ring (because u is a unit and $\chi^\sharp$ is a ring homomorphism). The units of $\mathbb{C}[t_1^{\pm 1}, \dots, t_n^{\pm 1}]$ are precisely the scalar multiples of Laurent monomials, $f = c\, t_1^{m_1} \cdots t_n^{m_n}$ with $c \in \mathbb{C}^*$ and $m \in \mathbb{Z}^n$; this description of the unit group of the Laurent polynomial ring is a fact of commutative algebra, which is proved in [2]. Thus $\chi(t) = c\, t_1^{m_1} \cdots t_n^{m_n}$ as a function. It remains to see $c = 1$. Since χ is a group homomorphism, it sends the identity $\mathbf{1} = (1, \dots, 1)$ of $(\mathbb{C}^*)^n$ to the identity 1 of $\mathbb{C}^*$; but $\chi(\mathbf{1}) = c \cdot 1 = c$, so $c = 1$. Therefore $\chi = \chi^m$, and combined with Steps 1 and 2 the map $m \mapsto \chi^m$ is a group isomorphism $\mathbb{Z}^n \xrightarrow{\sim} M$, giving $M \cong \mathbb{Z}^n$.

2. *Step 1: The monomial curves are one-parameter subgroups.* Fix $u = (u_1, \dots, u_n) \in \mathbb{Z}^n$ and set $\lambda^u(t) = (t^{u_1}, \dots, t^{u_n})$. Each coordinate t^{u_i} is a Laurent monomial in the single variable $t \in \mathbb{C}^*$, hence a unit of $\mathbb{C}[t^{\pm 1}]$, so λ^u maps $\mathbb{C}^*$ into $(\mathbb{C}^*)^n$ and is a morphism of varieties. It is a homomorphism because in each coordinate $(st)^{u_i} = s^{u_i} t^{u_i}$, so

$$\lambda^u(st) = \big((st)^{u_1}, \dots, (st)^{u_n}\big) = \big(s^{u_1} t^{u_1}, \dots, s^{u_n} t^{u_n}\big) = \lambda^u(s)\, \lambda^u(t)$$

the last product being the component-wise multiplication of $(\mathbb{C}^*)^n$. Thus λ^u is a toric morphism.

Step 2: Distinct exponent vectors give distinct one-parameter subgroups. If $\lambda^u = \lambda^{u'}$, then reading off the i-th coordinate gives $t^{u_i} = t^{u_i'}$ for all $t \in \mathbb{C}^*$, whence $u_i = u_i'$ by the same argument as in part (1), Step 2. So $u = u'$, and $u \mapsto \lambda^u$ is injective; it is a homomorphism because $\lambda^u \lambda^{u'} = \lambda^{u+u'}$ coordinatewise.

Step 3: Every one-parameter subgroup is a monomial curve. Let $\lambda\colon \mathbb{C}^* \to (\mathbb{C}^*)^n$ be a one-parameter subgroup. Composing with the i-th coordinate projection $\mathrm{pr}_i\colon (\mathbb{C}^*)^n \to \mathbb{C}^*$, which is itself a character, yields a character $\mathrm{pr}_i \circ \lambda\colon \mathbb{C}^* \to \mathbb{C}^*$ of the one-dimensional torus $\mathbb{C}^*$. By part (1) applied to $n = 1$, every character of $\mathbb{C}^*$ is $t \mapsto t^{u_i}$ for a unique $u_i \in \mathbb{Z}$. Hence the i-th coordinate of $\lambda(t)$ is t^{u_i}, and assembling the coordinates gives $\lambda(t) = (t^{u_1}, \dots, t^{u_n}) = \lambda^u(t)$ for $u = (u_1, \dots, u_n) \in \mathbb{Z}^n$. Combined with Steps 1 and 2, $u \mapsto \lambda^u$ is a group isomorphism $\mathbb{Z}^n \xrightarrow{\sim} N$, giving $N \cong \mathbb{Z}^n$.

Both M and N are therefore isomorphic to $\mathbb{Z}^n$, free abelian of rank $n = \dim(\mathbb{C}^*)^n$, completing the proof.

The proof isolates the mechanism cleanly: a character is determined by where it sends the coordinate function u of $\mathbb{C}^*$, and that image is forced to be a unit of the Laurent ring, hence a monomial with unit coefficient; the group law then kills the coefficient. Everything reduces to the description of the units of $\mathbb{C}[t_1^{\pm 1}, \dots, t_n^{\pm 1}]$, which is the algebraic form of the geometric statement that $(\mathbb{C}^*)^n$ has no nonconstant invertible regular functions other than monomials. The cocharacter computation is not independent; it is the character computation for $\mathbb{C}^*$ applied coordinate by coordinate, which is why part (2) leans on part (1) with $n = 1$.

The identification $M \cong \mathbb{Z}^n$ and $N \cong \mathbb{Z}^n$ should not obscure a distinction that becomes central later. The isomorphisms depend on the chosen coordinates $t_1, \ldots, t_n$ on T; a different choice of coordinates (that is, a different identification $T \cong (\mathbb{C}^*)^n$) changes the two isomorphisms by dual integer matrices. For an abstract torus T there is no preferred basis of M or N, only the intrinsic lattices $\mathrm{Hom}(T, \mathbb{G}_m)$ and $\mathrm{Hom}(\mathbb{G}_m, T)$. This is the same phenomenon as a finite-dimensional vector space having a well-defined dimension but no canonical basis, and it is what makes the passage from tori to lattices an equivalence of categories rather than a mere counting statement. That equivalence, together with the pairing $M \times N \to \mathbb{Z}$ that makes N the dual of M, is developed in section 1.3.

One consequence is worth recording immediately, since it is used constantly: a character χ^m is the constant map 1 if and only if $m = 0$. Indeed $\chi^m \equiv 1$ means $t_1^{m_1} \cdots t_n^{m_n} = 1$ for all $t \in (\mathbb{C}^*)^n$, and setting all but the j-th coordinate to 1 forces $m_j = 0$ for each j. Thus distinct lattice points give distinct characters, a fact the toric dictionary relies on when it records a monomial by its exponent vector.

The examples below compute characters and cocharacters in several concrete tori, and illustrate a subtlety that the isomorphism $N \cong \mathbb{Z}^n$ can hide: a one-parameter subgroup is more than its image. Two different lattice points u can trace out the same subset of T at different speeds, so the map $\lambda^u \mapsto \mathrm{im}\lambda^u$ is far from injective.

Example 1.2.4: Characters and Cocharacters in Small Tori

1. *The one-dimensional torus* $\mathbb{C}^*$. Here $M = \mathbb{Z}$ and $N = \mathbb{Z}$. The character $\chi^m \colon \mathbb{C}^* \to \mathbb{C}^*$ is $t \mapsto t^m$, and the one-parameter subgroup $\lambda^u \colon \mathbb{C}^* \to \mathbb{C}^*$ is $t \mapsto t^u$. For the one-dimensional torus characters and cocharacters coincide as functions $t \mapsto t^k$; the distinction between M and N is invisible until $n \geq 2$, where they pair against each other rather than with themselves.

2. *The two-dimensional torus* $(\mathbb{C}^*)^2$. Now $M = \mathbb{Z}^2$ and $N = \mathbb{Z}^2$. Taking $m = (2, -3)$, the character is
$$\chi^{(2,-3)}(t_1, t_2) = t_1^2\, t_2^{-3} = \frac{t_1^2}{t_2^3}$$
a Laurent monomial with a negative exponent, well-defined because $t_2 \in \mathbb{C}^*$ is invertible. Taking $u = (1, 2)$, the one-parameter subgroup is
$$\lambda^{(1,2)}(t) = (t,\, t^2)$$
whose image is the curve $\{(s, s^2) \,\big|\, s \in \mathbb{C}^*\}$, a copy of $\mathbb{C}^*$ sitting diagonally inside $(\mathbb{C}^*)^2$. Evaluating the character along this cocharacter gives
$$\chi^{(2,-3)}\big(\lambda^{(1,2)}(t)\big) = t^2\,(t^2)^{-3} = t^{2-6} = t^{-4}$$
a character of $\mathbb{C}^*$ with exponent $-4 = (2, -3) \cdot (1, 2) = 2 \cdot 1 + (-3) \cdot 2$. The exponent is the dot product $\langle m, u \rangle$ of the two lattice points, which is precisely the pairing $M \times N \to \mathbb{Z}$ that section 1.3 makes into the basis of the theory.

3. *Distinct one-parameter subgroups with equal image.* Consider the two one-parameter subgroups of $(\mathbb{C}^*)^2$ given by $u = (1, 0)$ and $u' = (2, 0)$:
$$\lambda^{(1,0)}(t) = (t, 1), \qquad \lambda^{(2,0)}(t) = (t^2, 1)$$

These are *different* elements of $N = \mathbb{Z}^2$, since $(1,0) \neq (2,0)$, and they are different as functions: $\lambda^{(1,0)}(2) = (2,1)$ while $\lambda^{(2,0)}(2) = (4,1)$. Yet they have the *same image*. Indeed

$$\mathrm{im}\lambda^{(1,0)} = \{(s,1) \big| s \in \mathbb{C}^*\} = \{(s^2,1) \big| s \in \mathbb{C}^*\} = \mathrm{im}\lambda^{(2,0)}$$

because squaring is a surjection $\mathbb{C}^* \to \mathbb{C}^*$ (every nonzero complex number has a square root). The two subgroups trace out the same subtorus $\mathbb{C}^* \times \{1\}$, but $\lambda^{(2,0)}$ winds around it "twice as fast": it is the composite of $\lambda^{(1,0)}$ with the double-cover $t \mapsto t^2$ of $\mathbb{C}^*$. A one-parameter subgroup is thus a *parametrized* curve in T, remembering both the subtorus and the speed of traversal, and $\lambda^u \mapsto \mathrm{im}\lambda^u$ forgets the speed.

The third computation is the reason the cocharacter lattice is defined as a group of homomorphisms and not as a set of subtori. The lattice point u carries strictly more information than the subgroup $\mathrm{im}\lambda^u$: it records the parametrization. This distinction will matter when one-parameter subgroups are used to probe the geometry of a toric variety by taking limits $\lim_{t\to 0} \lambda^u(t)$, where the direction of u (and not just the subtorus it generates) determines which boundary point the curve approaches. For now the point to retain is the clean bookkeeping the proposition provides: characters are exponent vectors in $M \cong \mathbb{Z}^n$, one-parameter subgroups are exponent vectors in $N \cong \mathbb{Z}^n$, and both are read off coordinate by coordinate from the standard torus.

Exercise 1.2.1

1. Write out the character χ^m of $(\mathbb{C}^*)^3$ for $m = (3,0,-2)$ explicitly as a Laurent monomial, and the one-parameter subgroup λ^u for $u = (-1,4,1)$. Compute the composite character $\chi^m \circ \lambda^u \colon \mathbb{C}^* \to \mathbb{C}^*$ and identify its exponent as an integer dot product.

2. Let χ, χ' be characters of a torus T. Prove directly from definition 1.2.1 that the pointwise product $\chi \cdot \chi'$ and the pointwise inverse $t \mapsto \chi(t)^{-1}$ are again characters, so that $M = \mathrm{Hom}(T, \mathbb{G}_m)$ is a group. Where is the fact that $\mathbb{G}_m$ is abelian used?

3. Show that the coordinate projection $\mathrm{pr}_i \colon (\mathbb{C}^*)^n \to \mathbb{C}^*$, $t \mapsto t_i$, is a character, and identify the lattice point $m \in \mathbb{Z}^n$ with $\mathrm{pr}_i = \chi^m$. Conclude that the characters $\mathrm{pr}_1, \ldots, \mathrm{pr}_n$ form a $\mathbb{Z}$-basis of M.

4. Prove that a character χ^m of $(\mathbb{C}^*)^n$ is a constant function if and only if $m = 0$, and that its image is all of $\mathbb{C}^*$ whenever $m \neq 0$.

5. Find all one-parameter subgroups λ^u of $(\mathbb{C}^*)^2$ whose image is the subtorus $\{(s,s) \big| s \in \mathbb{C}^*\}$ (the diagonal). Among them, which is the one whose exponent vector u is not a proper multiple of any other integer vector, and why is it the "slowest" traversal?

6. Let λ^u and $\lambda^{u'}$ be one-parameter subgroups of $(\mathbb{C}^*)^n$ with $u, u' \neq 0$. Show that $\mathrm{im}\lambda^u = \mathrm{im}\lambda^{u'}$ if and only if u and u' span the same line, that is, $u' = \frac{a}{b}u$ for some nonzero integers a, b (equivalently $\mathbb{Q}u = \mathbb{Q}u'$ inside $\mathbb{Q}^n$). Deduce that $u \neq u'$ can still give equal images, and give an example beyond $\{(1,0),(2,0)\}$ of example 1.2.4. Contrast this with the fact that λ^u and $\lambda^{u'}$ are equal *as maps* if and only if $u = u'$.

1.3 The Duality Pairing and the Lattice Dictionary

The previous section produced two lattices attached to a torus T: the character lattice $M = \mathrm{Hom}(T, \mathbb{G}_m)$ (definition 1.2.1) and the cocharacter lattice $N = \mathrm{Hom}(\mathbb{G}_m, T)$ (definition 1.2.2), each free abelian of rank $n = \dim T$ (proposition 1.2.3). These two lattices share more than a common rank; they are dual to one another. A character and a one-parameter subgroup can be composed, and the composite is again a homomorphism $\mathbb{G}_m \to \mathbb{G}_m$, which by proposition 1.2.3 is $t \mapsto t^k$ for a single integer k. That integer is the value of a bilinear pairing between M and N, and this pairing is perfect: it identifies each lattice with the $\mathbb{Z}$-linear dual of the other. Once this is in hand the entire correspondence built so far can be inverted. A lattice already carries enough data to reconstruct a torus, by tensoring with $\mathbb{C}^*$, and a homomorphism of tori is nothing more than a $\mathbb{Z}$-linear map of the underlying lattices. The passage from a torus to its cocharacter lattice and back is an equivalence of categories. This is the first and most basic instance of the combinatorics↔geometry dictionary that organizes the entire book, and it is what makes the subject combinatorial from the outset: a question about tori and toric morphisms is, verbatim, a question about lattices and integer matrices.

Definition 1.3.1: The Duality Pairing

Let T be a torus with character lattice M and cocharacter lattice N. For $m \in M$ and $u \in N$, the composite

$$\chi^m \circ \lambda^u \colon \mathbb{G}_m \xrightarrow{\lambda^u} T \xrightarrow{\chi^m} \mathbb{G}_m$$

is a homomorphism of algebraic groups $\mathbb{G}_m \to \mathbb{G}_m$, hence of the form $t \mapsto t^k$ for a unique integer k (proposition 1.2.3). This integer is denoted $\langle m, u\rangle$, so that

$$(\chi^m \circ \lambda^u)(t) = t^{\langle m,u\rangle} \tag{1.1}$$

The resulting map $\langle -, -\rangle \colon M \times N \to \mathbb{Z}$ is the *duality pairing* of T.

The pairing is defined without any choice of coordinates, purely from the group structure. To compute with it, fix an isomorphism $T \cong (\mathbb{C}^*)^n$, so that $M \cong \mathbb{Z}^n$ and $N \cong \mathbb{Z}^n$ by proposition 1.2.3: a character is $\chi^m(t) = t_1^{a_1} \cdots t_n^{a_n}$ for $m = (a_1, \dots, a_n)$, and a one-parameter subgroup is $\lambda^u(s) = (s^{b_1}, \dots, s^{b_n})$ for $u = (b_1, \dots, b_n)$. Substituting one into the other,

$$(\chi^m \circ \lambda^u)(s) = \chi^m(s^{b_1}, \dots, s^{b_n}) = (s^{b_1})^{a_1} \cdots (s^{b_n})^{a_n} = s^{a_1 b_1 + \cdots + a_n b_n}$$

so that in coordinates the pairing is the standard dot product

$$\langle m, u\rangle = \sum_{i=1}^{n} a_i b_i \tag{1.2}$$

The pairing is $\mathbb{Z}$-bilinear: this is immediate from (1.2), and can also be read off intrinsically, since $\chi^{m+m'} = \chi^m \chi^{m'}$ makes $\langle -, u\rangle$ additive in the first slot and $\lambda^{u+u'} = \lambda^u \lambda^{u'}$ makes $\langle m, -\rangle$ additive in the second (here the products are taken pointwise in T, and the second identity uses that T is abelian). The next example computes the pairing on a running fixture and records the intrinsic bilinearity computation in coordinates.

Example 1.3.2: The Standard Pairing on $(\mathbb{C}^*)^2$

Take $T = (\mathbb{C}^*)^2$, with $M = \mathbb{Z}^2$ and $N = \mathbb{Z}^2$ under the identifications of proposition 1.2.3. Let $m = (3, -2) \in M$ and $u = (1, 4) \in N$, so that

$$\chi^m(t_1, t_2) = t_1^3 t_2^{-2}, \qquad \lambda^u(s) = (s^1, s^4)$$

Composing gives $(\chi^m \circ \lambda^u)(s) = (s^1)^3(s^4)^{-2} = s^{3-8} = s^{-5}$, so $\langle m, u\rangle = -5$, in agreement with $3 \cdot 1 + (-2) \cdot 4 = -5$ from (1.2). Taking instead the standard bases $e_1 = (1, 0), e_2 = (0, 1)$ of M and $f_1 = (1, 0), f_2 = (0, 1)$ of N, one finds

$$\langle e_i, f_j\rangle = \delta_{ij}$$

the Kronecker delta. Thus $\{e_i\}$ and $\{f_j\}$ are *dual bases*: the pairing matrix is the identity. This is the reason M and N are drawn as "the same" $\mathbb{Z}^2$ in coordinates while being kept conceptually distinct, one recording monomials, the other recording directions of approach.

The Kronecker relation $\langle e_i, f_j\rangle = \delta_{ij}$ in example 1.3.2 is the crux: it says the matrix of the pairing in dual bases is invertible over $\mathbb{Z}$. Spelled out, this is the assertion that the pairing is *perfect*, which is the property that lets each lattice be recovered as the dual of the other.

Proposition 1.3.3: The Duality Pairing is Perfect

Let T be a torus with lattices M and N and duality pairing $\langle -, -\rangle$. The two induced $\mathbb{Z}$-linear maps

$$M \longrightarrow \mathrm{Hom}(N, \mathbb{Z}), \quad m \mapsto \langle m, -\rangle, \qquad N \longrightarrow \mathrm{Hom}(M, \mathbb{Z}), \quad u \mapsto \langle -, u\rangle$$

are isomorphisms. Consequently $M = \mathrm{Hom}(N, \mathbb{Z})$ and $N = \mathrm{Hom}(M, \mathbb{Z})$ canonically.

***Proof*:**

It suffices to treat the first map, as the second is the same statement with the roles of M and N exchanged: the coordinate formula (1.2) is symmetric in its two arguments, so the second map is the first with the roles of M and N interchanged. Fix an isomorphism $T \cong (\mathbb{C}^*)^n$, giving dual bases $e_1, \ldots, e_n$ of M and $f_1, \ldots, f_n$ of N with $\langle e_i, f_j\rangle = \delta_{ij}$, exactly as computed in example 1.3.2 (the computation there for $n = 2$ is identical in general, since $\chi^{e_i}(\lambda^{f_j}(s)) = s^{\delta_{ij}}$).

Let $f_1^*, \ldots, f_n^*$ be the basis of $\mathrm{Hom}(N, \mathbb{Z})$ dual to $f_1, \ldots, f_n$, so $f_i^*(f_j) = \delta_{ij}$. The map $m \mapsto \langle m, -\rangle$ sends e_i to the functional $u \mapsto \langle e_i, u\rangle$, which on the basis element f_j takes the value $\langle e_i, f_j\rangle = \delta_{ij} = f_i^*(f_j)$. A $\mathbb{Z}$-linear functional is determined by its values on a basis, so $\langle e_i, -\rangle = f_i^*$. The map therefore carries the basis $\{e_i\}$ of M bijectively to the basis $\{f_i^*\}$ of $\mathrm{Hom}(N, \mathbb{Z})$, and a $\mathbb{Z}$-linear map carrying a basis to a basis is an isomorphism, as we sought to show.

proposition 1.3.3 is the formal content of the phrase "M and N are dual lattices." It has a concrete payoff: any $\mathbb{Z}$-linear functional $N \to \mathbb{Z}$ is $\langle m, -\rangle$ for a unique $m \in M$, so a linear condition on cocharacters is the same datum as an element of M. This is exactly how, in **??**, a supporting hyperplane of a cone in $N_\mathbb{R}$ will be recorded by a lattice point of M, and it is why the dual cone $\sigma^\vee$ lives in $M_\mathbb{R}$ while σ lives in $N_\mathbb{R}$. For those later geometric arguments the pairing must be available over $\mathbb{R}$, not only over $\mathbb{Z}$, so that half-spaces and hyperplanes make sense. Since M and N are free, extending scalars is harmless.

Definition 1.3.4: The Real Pairing

Write $M_{\mathbb{R}} = M \otimes_{\mathbb{Z}} \mathbb{R}$ and $N_{\mathbb{R}} = N \otimes_{\mathbb{Z}} \mathbb{R}$, the real vector spaces spanned by M and N; each has dimension $n = \operatorname{rank} M = \operatorname{rank} N$. Extending $\langle -, - \rangle$ $\mathbb{R}$-bilinearly in each variable produces the *real pairing*

$$\langle -, - \rangle \colon M_{\mathbb{R}} \times N_{\mathbb{R}} \to \mathbb{R}, \qquad \Big\langle \sum_i r_i m_i,\ \sum_j s_j u_j \Big\rangle = \sum_{i,j} r_i s_j \langle m_i, u_j \rangle$$

for $r_i, s_j \in \mathbb{R}$, $m_i \in M$, $u_j \in N$. It is a perfect pairing of real vector spaces, identifying $M_{\mathbb{R}}$ with the dual space $(N_{\mathbb{R}})^*$ and $N_{\mathbb{R}}$ with $(M_{\mathbb{R}})^*$.

The real pairing is perfect for the same reason as proposition 1.3.3: in dual bases its matrix is the identity, now over $\mathbb{R}$. In coordinates it is again the dot product (1.2), extended to real arguments. Nothing new happens; the extension only widens the coefficients so that convex-geometric objects, which need real scalars, have somewhere to live. With the pairing established the perspective can be reversed. Up to now a torus was given and its lattices extracted; the next result shows a lattice suffices to build the torus.

Before stating it, recall why tensoring with $\mathbb{C}^*$ is the right construction. A torus $(\mathbb{C}^*)^n$ is n independent copies of $\mathbb{C}^*$, one for each basis direction of N; abstractly, choosing a copy of $\mathbb{C}^*$ for each basis vector of N and remembering the $\mathbb{Z}$-linear combinations is precisely the tensor product $N \otimes_{\mathbb{Z}} \mathbb{C}^*$. The tensor product makes this coordinate-free, so that no basis of N is privileged.

Proposition 1.3.5: The Torus of a Lattice

Let N be a lattice, that is, a free abelian group of finite rank n. Define

$$T_N := N \otimes_{\mathbb{Z}} \mathbb{C}^*$$

where $\mathbb{C}^* = \mathbb{G}_m$ is regarded as an abelian group under multiplication. Then:

1. A choice of $\mathbb{Z}$-basis $u_1, \ldots, u_n$ of N determines an isomorphism $T_N \cong (\mathbb{C}^*)^n$ of algebraic groups, sending $\sum_i u_i \otimes t_i$ to $(t_1, \ldots, t_n)$; thus T_N is a torus of dimension n.
2. The cocharacter lattice of T_N is canonically N, via $u \mapsto \lambda^u$ where $\lambda^u(t) = u \otimes t$.
3. The character lattice of T_N is canonically $M = \operatorname{Hom}(N, \mathbb{Z})$, via $m \mapsto \chi^m$ where $\chi^m(u \otimes t) = t^{\langle m, u \rangle}$, extended multiplicatively; and the duality pairing of T_N recovers the evaluation pairing $M \times N \to \mathbb{Z}$.

***Proof*:**
Step 1: The tensor product and the coordinate isomorphism. Since N is free with basis $u_1, \ldots, u_n$, there is a group isomorphism $N \cong \mathbb{Z}^n$, and tensoring the (right-exact, indeed exact on free modules) functor $- \otimes_{\mathbb{Z}} \mathbb{C}^*$ gives

$$T_N = N \otimes_{\mathbb{Z}} \mathbb{C}^* \cong \mathbb{Z}^n \otimes_{\mathbb{Z}} \mathbb{C}^* \cong (\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{C}^*)^n \cong (\mathbb{C}^*)^n$$

where the middle isomorphism distributes the tensor over the direct sum $\mathbb{Z}^n = \bigoplus_i \mathbb{Z} u_i$, and the last uses $\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{C}^* \cong \mathbb{C}^*$. Tracking a simple tensor through these identifications, $u_i \otimes t \mapsto e_i \otimes t \mapsto (1, \ldots, t, \ldots, 1)$ with t in slot i, so a general element $\sum_i u_i \otimes t_i$ maps to $(t_1, \ldots, t_n)$. This is a bijection with inverse $(t_1, \ldots, t_n) \mapsto \sum_i u_i \otimes t_i$. It is a group homomorphism because addition in the tensor product corresponds, under $u_i \otimes t_i \cdot u_i \otimes t_i' = u_i \otimes t_i t_i'$, to coordinatewise multiplication in $(\mathbb{C}^*)^n$. Transporting the variety structure of $(\mathbb{C}^*)^n$ (example 1.1.5) across this bijection makes T_N an algebraic group isomorphic to $(\mathbb{C}^*)^n$, hence a torus of dimension n (definition 1.1.3). A different basis $u_1', \ldots, u_n'$ is related to $u_1, \ldots, u_n$ by a matrix in $\mathrm{GL}_n(\mathbb{Z})$, whose entries give an isomorphism $(\mathbb{C}^*)^n \to (\mathbb{C}^*)^n$ of algebraic groups (Laurent monomials in the coordinates), so the torus structure on T_N is independent of the basis chosen.

Step 2: The cocharacter lattice. Fix the basis $u_1, \ldots, u_n$ and the resulting isomorphism $T_N \cong (\mathbb{C}^*)^n$. For $u \in N$ define $\lambda^u \colon \mathbb{G}_m \to T_N$ by $\lambda^u(t) = u \otimes t$; this is a homomorphism of algebraic groups since $u \otimes (tt') = u \otimes t \cdot u \otimes t'$ and, writing $u = \sum_i c_i u_i$ with $c_i \in \mathbb{Z}$, in coordinates $\lambda^u(t) = (t^{c_1}, \ldots, t^{c_n})$, a one-parameter subgroup. The assignment $u \mapsto \lambda^u$ is a group homomorphism $N \to \mathrm{Hom}(\mathbb{G}_m, T_N)$ because $\lambda^{u+u'}(t) = (u+u') \otimes t = u \otimes t \cdot u' \otimes t = \lambda^u(t)\lambda^{u'}(t)$. By proposition 1.2.3 every one-parameter subgroup of $(\mathbb{C}^*)^n$ has the form $t \mapsto (t^{c_1}, \ldots, t^{c_n})$ for a unique $(c_1, \ldots, c_n) \in \mathbb{Z}^n$, which is exactly λ^u for $u = \sum_i c_i u_i$; hence $u \mapsto \lambda^u$ is a bijection, an isomorphism $N \xrightarrow{\sim} \mathrm{Hom}(\mathbb{G}_m, T_N)$ of lattices. Under the coordinate identification $N \cong \mathbb{Z}^n$ this is the identity, so the cocharacter lattice of T_N is canonically N.

Step 3: The character lattice and the pairing. Let $m \in M = \mathrm{Hom}(N, \mathbb{Z})$. Define $\chi^m \colon T_N \to \mathbb{G}_m$ on simple tensors by $\chi^m(u \otimes t) = t^{\langle m,u \rangle}$, where $\langle m, u\rangle = m(u)$ is evaluation, and extend to all of T_N multiplicatively. To see this is well defined and a homomorphism, pass to coordinates: writing m in the basis $u_1^*, \ldots, u_n^*$ of M dual to $u_1, \ldots, u_n$ as $m = \sum_i a_i u_i^*$, evaluation gives $\langle m, u_i \rangle = a_i$, and on a general point $\sum_i u_i \otimes t_i \leftrightarrow (t_1, \ldots, t_n)$,

$$\chi^m\Big(\sum_i u_i \otimes t_i\Big) = \prod_i t_i^{\langle m, u_i\rangle} = t_1^{a_1} \cdots t_n^{a_n}$$

which is the Laurent monomial character of $(\mathbb{C}^*)^n$ attached to $(a_1, \ldots, a_n) \in \mathbb{Z}^n$. By proposition 1.2.3 every character of $(\mathbb{C}^*)^n$ arises this way from a unique such tuple, so $m \mapsto \chi^m$ is a bijection $M \xrightarrow{\sim} \mathrm{Hom}(T_N, \mathbb{G}_m)$; it is a group homomorphism because $\chi^{m+m'} = \chi^m \chi^{m'}$ follows from $t^{\langle m+m', u\rangle} = t^{\langle m,u\rangle} t^{\langle m',u\rangle}$. Thus the character lattice of T_N is canonically $M = \mathrm{Hom}(N, \mathbb{Z})$. Finally, the duality pairing of T_N evaluates $\chi^m \circ \lambda^u$: from $(\chi^m \circ \lambda^u)(t) = \chi^m(u \otimes t) = t^{\langle m,u\rangle}$ its value is $\langle m, u\rangle = m(u)$, the evaluation pairing, completing the proof.

proposition 1.3.5 closes the loop opened in section 1.2. There a torus produced a pair of dual lattices; here a lattice produces a torus whose lattices are the one started with. The two constructions, $T \mapsto N$ and $N \mapsto T_N$, are visibly inverse to each other on objects. The content of the lattice dictionary is that they are inverse on *morphisms* as well, and this requires understanding how a toric morphism moves through the correspondence. A homomorphism of tori acts on one-parameter subgroups by composition, and this action is the promised $\mathbb{Z}$-linear map of cocharacter lattices.

Theorem 1.3.6: The Lattice Dictionary

Let **Tori** be the category whose objects are tori and whose morphisms are toric morphisms (definition 1.1.4), and let **Lat** be the category of lattices with $\mathbb{Z}$-linear maps. The assignments

$$N \longmapsto T_N, \qquad T \longmapsto N_T := \mathrm{Hom}(\mathbb{G}_m, T)$$

extend to mutually inverse equivalences of categories $\mathbf{Lat} \rightleftarrows \mathbf{Tori}$. Concretely:

1. A toric morphism $\varphi\colon T \to T'$ induces a $\mathbb{Z}$-linear map $\varphi_*\colon N_T \to N_{T'}$ by $\varphi_*(\lambda) = \varphi \circ \lambda$, and every $\mathbb{Z}$-linear map $\psi\colon N_T \to N_{T'}$ is φ_* for a unique toric morphism φ. Thus

$$\mathrm{Hom}_{\mathbf{Tori}}(T, T') \cong \mathrm{Hom}_{\mathbb{Z}}(N_T, N_{T'})$$

is a bijection, natural in T and T'.

2. Dually, φ induces $\widehat{\varphi}\colon M' \to M$ on character lattices by $\widehat{\varphi}(\chi) = \chi \circ \varphi$, and $\widehat{\varphi} = (\varphi_*)^\vee$ is the transpose of φ_* with respect to the duality pairings: $\langle \widehat{\varphi}(m'), u\rangle = \langle m', \varphi_*(u)\rangle$ for all $m' \in M'$, $u \in N_T$.

***Proof*:**
Step 1: Functoriality of $N_{(-)}$. For a toric morphism $\varphi\colon T \to T'$ and $\lambda \in N_T = \mathrm{Hom}(\mathbb{G}_m, T)$, the composite $\varphi \circ \lambda\colon \mathbb{G}_m \to T'$ is a morphism of varieties and a group homomorphism (both φ and λ are), so $\varphi \circ \lambda \in N_{T'}$. The map $\varphi_*(\lambda) = \varphi \circ \lambda$ is $\mathbb{Z}$-linear: for $\lambda, \mu \in N_T$ the sum $\lambda + \mu$ in N_T is the pointwise product $t \mapsto \lambda(t)\mu(t)$ in T, and since φ is a group homomorphism, $\varphi(\lambda(t)\mu(t)) = \varphi(\lambda(t))\varphi(\mu(t))$, so $\varphi_*(\lambda + \mu) = \varphi_*(\lambda) + \varphi_*(\mu)$. Functoriality is immediate: $(\psi \circ \varphi)_* = \psi_* \circ \varphi_*$ from associativity of composition, and $(\mathrm{id}_T)_* = \mathrm{id}_{N_T}$.

Step 2: The map $\varphi \mapsto \varphi_*$ is a bijection on hom-sets. Fix bases, identifying $T \cong (\mathbb{C}^*)^n$ and $T' \cong (\mathbb{C}^*)^{n'}$, so $N_T \cong \mathbb{Z}^n$ and $N_{T'} \cong \mathbb{Z}^{n'}$. A toric morphism $\varphi\colon (\mathbb{C}^*)^n \to (\mathbb{C}^*)^{n'}$ is a group homomorphism given by morphisms of varieties in each coordinate; each coordinate $\varphi_j\colon (\mathbb{C}^*)^n \to \mathbb{C}^*$ is itself a character of $(\mathbb{C}^*)^n$ (a variety morphism to $\mathbb{C}^*$ that is a group homomorphism), hence by proposition 1.2.3 a Laurent monomial $\varphi_j(t) = t_1^{a_{j1}} \cdots t_n^{a_{jn}}$ with $a_{ji} \in \mathbb{Z}$. Therefore φ is recorded exactly by the integer matrix $A = (a_{ji}) \in \mathrm{Mat}_{n' \times n}(\mathbb{Z})$, and conversely every such matrix defines a toric morphism $t \mapsto (\prod_i t_i^{a_{1i}}, \ldots, \prod_i t_i^{a_{n'i}})$. Applying φ to the one-parameter subgroup λ^u, $u = (c_1, \ldots, c_n)$, gives $(\varphi \circ \lambda^u)(s) = \varphi(s^{c_1}, \ldots, s^{c_n})$, whose j-th coordinate is $\prod_i (s^{c_i})^{a_{ji}} = s^{\sum_i a_{ji}c_i} = s^{(Au)_j}$, so $\varphi_*(\lambda^u) = \lambda^{Au}$. Under $N_T \cong \mathbb{Z}^n$, $N_{T'} \cong \mathbb{Z}^{n'}$ the map φ_* is therefore multiplication by the same matrix A. Since $\varphi \leftrightarrow A \leftrightarrow \varphi_*$ are all recorded by the identical matrix, $\varphi \mapsto \varphi_*$ is a bijection $\mathrm{Hom}_{\mathbf{Tori}}(T, T') \to \mathrm{Hom}_{\mathbb{Z}}(N_T, N_{T'})$; a $\mathbb{Z}$-linear map ψ corresponds to its matrix, which reconstructs the unique φ. Naturality in T and T' follows because composing with a further toric morphism corresponds to matrix multiplication on either side.

Step 3: The two functors are mutually inverse. proposition 1.3.5(2) gives a canonical isomorphism $N \cong N_{T_N}$, natural in N: a $\mathbb{Z}$-linear map $\psi\colon N \to N'$ has matrix A in chosen bases, and the toric morphism $T_N \to T_{N'}$ it corresponds to under Step 2 is $x \mapsto$ (the Laurent-monomial map with matrix A), whose induced map on cocharacters is again A, so the square relating ψ and $(T_\psi)_*$

commutes. In the other direction, for a torus T the composite $T \rightsquigarrow N_T \rightsquigarrow T_{N_T}$ returns T up to the canonical isomorphism $T \cong T_{N_T}$ sending $\lambda \otimes t \mapsto \lambda(t)$: this is a well-defined homomorphism because $N_T \otimes \mathbb{C}^* \to T$, $\lambda \otimes t \mapsto \lambda(t)$, is bilinear, and in coordinates $\lambda^u \otimes t \mapsto (t^{c_1}, \dots, t^{c_n})$ realizes the isomorphism $T_{N_T} \cong (\mathbb{C}^*)^n \cong T$ of proposition 1.3.5(1). These isomorphisms are natural in N and in T respectively by the matrix bookkeeping of Step 2. Two functors with natural isomorphisms $N \cong N_{T_N}$ and $T \cong T_{N_T}$ to the respective identities are mutually inverse equivalences, which establishes the claimed equivalence and part (1).

Step 4: The dual map and the transpose identity. For $\varphi\colon T \to T'$ and a character $\chi \in M' = \operatorname{Hom}(T', \mathbb{G}_m)$, the composite $\chi \circ \varphi\colon T \to \mathbb{G}_m$ is a variety morphism and a group homomorphism, so lies in M; the assignment $\widehat{\varphi}(\chi) = \chi \circ \varphi$ is $\mathbb{Z}$-linear by the same pointwise-product argument as in Step 1, giving $\widehat{\varphi}\colon M' \to M$. To identify $\widehat{\varphi}$ with the transpose of φ_*, evaluate the pairing. For $m' \in M'$ and $u \in N_T$, using the definition (1.1) of the pairing and associativity of composition,

$$t^{\langle \widehat{\varphi}(m'), u\rangle} = (\chi^{\widehat{\varphi}(m')} \circ \lambda^u)(t) = \big((\chi^{m'} \circ \varphi) \circ \lambda^u\big)(t) = \big(\chi^{m'} \circ (\varphi \circ \lambda^u)\big)(t) = (\chi^{m'} \circ \lambda^{\varphi_*(u)})(t) = t^{\langle m', \varphi_*(u)\rangle}$$

for all $t \in \mathbb{G}_m$, whence $\langle \widehat{\varphi}(m'), u\rangle = \langle m', \varphi_*(u)\rangle$. This is precisely the statement that $\widehat{\varphi}$ is the transpose $(\varphi_*)^\vee$ under the perfect pairings of proposition 1.3.3; in dual bases, if φ_* has matrix A then $\widehat{\varphi}$ has matrix A^T, as we sought to show.

The dictionary is the organizing principle of the subject, and it deserves to be stated plainly: *the category of tori is the same as the category of lattices.* A torus, an intrinsically geometric object (a variety with a group law), is completely encoded by a finitely generated free abelian group, a piece of pure combinatorics, and a homomorphism of tori is completely encoded by an integer matrix. Nothing is lost in either direction. theorem 1.3.6 turns questions about tori into questions about lattices, which are questions about integer linear algebra, and the answers translate back without residue. A subtorus corresponds to a saturated sublattice, a quotient torus to a quotient lattice, and the dimension of a torus is the rank of its lattice; each of these is a routine consequence of the equivalence, worked out in the exercises.

This is the reason toric geometry is combinatorial. Every construction to come rests on this first translation. In **??** a strongly convex rational cone $\sigma \subseteq N_\mathbb{R}$ carves out a subsemigroup $S_\sigma = \sigma^\vee \cap M$ of M, and the affine toric variety U_σ is assembled from that combinatorial datum, its torus being T_N for the same N in which σ sits. The pairing of definition 1.3.1, extended to $M_\mathbb{R} \times N_\mathbb{R} \to \mathbb{R}$ in definition 1.3.4, is what lets σ in $N_\mathbb{R}$ and $\sigma^\vee$ in $M_\mathbb{R}$ speak to each other. That M and N are dual lattices, not just abstractly isomorphic, is what makes the cone and its dual carry the same information viewed from opposite sides, precisely the duality the whole book exploits. The lattice dictionary is the ground floor; everything is built on it.

Exercise 1.3.1

1. Let $T = (\mathbb{C}^*)^3$ with $M = \mathbb{Z}^3 = N$ under the standard identifications. Compute $\langle m, u\rangle$ for $m = (2, -1, 4)$ and $u = (3, 5, -2)$ directly from the definition, by writing out $\chi^m \circ \lambda^u$ as $t \mapsto t^k$ and reading off k. Then verify the answer against (1.2).
2. Consider the toric morphism $\varphi\colon (\mathbb{C}^*)^2 \to (\mathbb{C}^*)^2$ given by $\varphi(t_1, t_2) = (t_1^2 t_2,\ t_1 t_2^{-1})$. Write down the matrix A recording φ as in the proof of theorem 1.3.6, compute the induced map $\varphi_*\colon N \to N$ on cocharacters and $\widehat{\varphi}\colon M \to M$ on characters, and confirm that the two matrices are transposes of each other. Is φ an isomorphism of tori? Justify your answer using the

dictionary.

3. Prove that a toric morphism $\varphi\colon T \to T'$ is an isomorphism of tori if and only if the induced map $\varphi_*\colon N_T \to N_{T'}$ is an isomorphism of lattices, if and only if φ_* is represented (in some, equivalently any, bases) by a matrix in $\mathrm{GL}_n(\mathbb{Z})$. Deduce that two tori are isomorphic if and only if they have the same dimension.

4. Let $\varphi\colon T \to T'$ be a toric morphism with $\varphi_*\colon N_T \to N_{T'}$ injective, and let $\widehat{\varphi}\colon M' \to M$ be the dual map. Show that:

    1. the image $\widehat{\varphi}(M')$ always has finite index in M, so the sharp question is when $\widehat{\varphi}$ is *surjective* (index 1);
    2. $\widehat{\varphi}$ is surjective if and only if φ_* is a split injection (that is, $\varphi_*(N_T)$ is a saturated sublattice, a direct summand of $N_{T'}$);
    3. $\widehat{\varphi}$ is injective if and only if φ_* is injective with finite cokernel, that is, a finite-index inclusion of full-rank lattices.

    In particular, injectivity of φ_* alone already makes the image of $\widehat{\varphi}$ finite-index, whether or not φ_* is a bijection: the example $N_T = \mathbb{Z} \hookrightarrow \mathbb{Z}^2 = N_{T'}$, $a \mapsto (a, 0)$, has $\widehat{\varphi} = (1\ 0)\colon \mathbb{Z}^2 \to \mathbb{Z}$ surjective (image all of M) while φ_* is not a bijection. Interpret each condition geometrically in terms of φ.

5. Using proposition 1.3.5, show that for lattices N_1, N_2 there is a canonical isomorphism $T_{N_1 \oplus N_2} \cong T_{N_1} \times T_{N_2}$, and that the character lattice of a product of tori is the direct sum of the character lattices. What is the duality pairing of a product torus in terms of the pairings of the factors?

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